

pandas → data preprocessing library

import pandas as pd

df = pd.read_csv('file name')

df.head() df.tail()

df.columnname, df['column-name'] -

df.iloc[:, :] ✓

axis-0 df axis 1

mpg	displacement	(-)	(-)	(-)
18.0	8	-	-	-
15.0	7	-	-	-

df.shape[0] = 398 //

df.shape[1] = 9 //

df, df['horse power']

x Select all from df where df['horsepower'] == '?'

↳ df[df['horsepower'] == '?'] ✓

H	W	G
160	80	M
170	90	M
145	80	R
180	60	F
170	50	M

$$\underline{\underline{80 + 90 + 60 + 60}}$$

Steps to follow:

`df = df.read_csv('autompg.dat')` ✓

`df.dtypes` → horsepower → '?'

Convert to float

↳ `df['horsepower'] = df['horsepower'].astype(float)`

398

↓
we got error saying there is '?'
in the column

Option 1:

`df = df[df['horsepower'] != '?']`

Selecting all the rows without '?'

Selecting an ...

(392)

Option 2:

`df['horsepower'] = df['horsepower'].replace('?', 0.0)`

all the '?' replaced with '0.0'

(398)

Option 3:

`df['horsepower'] = df['horsepower'].replace('?', 'NaN')`

`mean-hp = df['horsepower'].mean()`

`df['horsepower'] = df['horsepower'].replace('NaN', mean-hp)`

(398) → replaced with average

Common

`df['horsepower'] = df['horsepower'].astype('float')`